Linear Approximation and L'Hôpital's Rule

Section 1: Tangent Line Approximation (Linearization)

1 What is the local linear approximation $L(x)$ of $f(x) = \sqrt{x}$ at $x = 25$?

a. $L(x) = 5 + \frac{1}{10}(x - 25)$

b. $L(x) = 5 + \frac{1}{5}(x - 25)$

c. $L(x) = 5 + \frac{1}{2}(x - 25)$

d. $L(x) = \frac{1}{10}x$

2 Using the linearization of $f(x) = \sqrt{x}$ at $a = 25$, approximate $\sqrt{26}$.

a. 5.10

b. 5.05

c. 5.20

d. 5.01

3 Is the tangent line approximation for $f(x) = \sqrt{x}$ an overestimate or underestimate, and why?

a. An overestimate because $f''(x) < 0$ (concave down)

b. An underestimate because $f''(x) > 0$ (concave up)

c. Exact because f is monotonic

d. Depends on the step size

Section 2: L'Hôpital's Rule for 0/0 Forms

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4 Evaluate $\lim_{x \to 0} \frac{e^{2x} - 1}{\sin(3x)}$.

a. $\frac{2}{3}$

b. $\frac{3}{2}$

c. 0

d. ∞

5 Evaluate $\lim_{x \to 1} \frac{\ln x}{x^2 - 1}$.

a. $\frac{1}{2}$

b. 1

c. 2

d. 0

6 Evaluate $\lim_{x \to 0} \frac{1 - \cos(4x)}{x^2}$.

a. 8

b. 4

c. 2

d. 16

Section 3: L'Hôpital's Rule for Infinity/Infinity Forms

7 Evaluate $\lim_{x \to \infty} \frac{\ln(x)}{x}$.

a. 0

b. 1

c. ∞

d. Undefined

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8 Evaluate $\lim_{x\to\infty} \frac{x^2}{e^x}$.

a. 0

b. 1

c. 2

d. ∞

9 Which requirement MUST be verified before applying L'Hôpital's Rule to $\lim_{x\to c} \frac{f(x)}{g(x)}$?

a. The limit produces an indeterminate form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$

b. $f(c) = 0$ only

c. $f'(c) = 0$

d. $g(x) > 0$ for all x

Section 4: Comparing Relative Growth Rates of Functions

10 As $x \to \infty$, which of the following functions grows the fastest?

a. e^x

b. x^{10}

c. $x\ln x$

d. 2^x

11 Evaluate $\lim_{x\to\infty} \frac{(\ln x)^5}{\sqrt{x}}$.

a. 0

b. ∞

c. 1

d. 5

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12 If $\lim_{x \to \infty} \frac{f(x)}{g(x)} = \infty$, what does this conclude about their growth rates?

a. $f(x)$ grows faster than $g(x)$

b. $g(x)$ grows faster than $f(x)$

c. They grow at the same rate

d. Neither function grows

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